GAUSSIAN PROCESS REGRESSION / A TEMPORAL STUDY

Forecasting Sri Lankan Recorded Dispatch Energy

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Abstract

This study evaluates one-day-ahead forecasting of recorded electricity dispatch energy in Sri Lanka using Gaussian Process Regression (GPR), a non-deep-learning algorithm absent from the supplied teaching material. The data are compiled from a versioned public release of PUCSL plant-level dispatch records. A raw-data audit identifies duplicate timestamp/plant keys and a published daily series that omits the documented interval conversion. Energy is therefore reconstructed from deduplicated 15-minute MW readings rather than accepted from the released daily CSV. Quality and lag-availability checks yield 756 modelling days: 659 for development and 97 for chronological testing.

Three expanding validation windows select a Matérn-5/2 kernel. On the frozen January-April 2025 test, GPR achieves MAE 1.599 GWh, RMSE 2.492 GWh and R^2 0.544, reducing MAE by 29.7% against a weekly seasonal-naïve forecast. SVR has lower test MAE and random forest lower RMSE; paired block resampling does not establish GPR superiority over random forest. Nominal 95% observation intervals cover 92.8% of outcomes. Grouped permutation importance and retrained ablations identify demand history and weekly calendar structure as useful, while local sensitivity analysis exposes reference dependence. Large errors coincide with several public holidays omitted from the frozen predictors. The resulting interface is a historical research prototype; limitations in data coverage, forecast-time availability and uncertainty prevent an operational grid-control claim.

A supplementary extension compares seven learned algorithms across electricity and an independent Colombo rainfall task. NGBoost is competitive on electricity but its rainfall intervals under-cover; the rainfall seasonal-median baseline outperforms all learned models. The main GPR study is preserved, with post-hoc electricity comparisons identified explicitly. A private source repository and an Azure Free F1 HTTPS application provide reviewable deployment evidence.

Keywords: Sri Lanka; dispatch energy; Gaussian processes; temporal validation; explainable machine learning; uncertainty.

Review status: AI-assisted research draft for student review and defence. Assistance and unresolved institutional-policy checks are disclosed in the final section. This report contains executed experimental results, rather than illustrative performance values.

1. Problem and local data

1.1 Research question and practical relevance

Can GPR improve a simple weekly forecast of the next local day's recorded dispatch energy, while supplying interpretable feature-reliance evidence and useful uncertainty information? Electricity planning benefits from anticipating daily energy requirements, but a forecasting study must distinguish measured dispatch from end-use consumption and from demand that might have been curtailed. This assignment investigates a carefully delimited dispatch proxy, rather than claiming to recover unconstrained national demand.

CEB reports average daily net generation of 45.9 GWh in 2024 [11]. The selected API subset reconstructs an average 42.136 GWh/day for that year. These are different measurement scopes; the discrepancy is a warning against relabelling this dataset as a complete national consumption census. The project's practical contribution is a reproducible local forecasting and data-quality study, supported by a demonstrable interface.

1.2 Collection and provenance

The study retrieves the Sri Lankan raw subset from Coombes, Hunt and Bloomfield's SASEA dataset, Zenodo record 17175212, version v1, published 22 September 2025 [2]. Its original producer is PUCSL. The compilers document retrieval of plant-level dispatch power through PUCSL's actual-system-dispatch API [2], [3]. The student's contribution is secondary acquisition, auditing, compilation and modelling, not original electricity measurement. Retrieval on 28 October 2026 is pinned by the archive MD5 and extracted-file SHA-256 in the provenance record. The verified licence is CC BY 4.0; attribution and changes are documented.

Raw field	Meaning	Unit / type
timestamp	UTC observation timestamp	ISO-8601
powerPlantId	Opaque generating-unit identifier	Integer
dispatchValueInMW	Recorded plant dispatch power	MW; may be null or negative

The raw file has 2,902,337 rows and 44 plant identifiers across the record. Its local timestamps run from 1 January 2023 at 05:30 to 8 April 2025 at 05:30, Asia/Colombo. The source YAML contains placeholder DOIs and inconsistent textual year ranges; measured timestamps and the minted Zenodo record resolve these discrepancies. The pinned upstream archive remains unchanged.

1.3 Ethical scope

The selected variables concern aggregate generating units, not identifiable households or patients. No personal or sensitive records are collected. Reuse still requires accurate attribution, faithful descriptions of coverage, and restraint about public policy or operational decisions. National aggregation cannot support individual-level fairness claims.

2. Reconstructing a defensible target

UTC timestamps are converted before deriving local dates. Duplicate timestamp/plant keys are checked for conflicting values: none are found. Removing 29,875 identical repeated keys leaves 79,489 distinct timestamps. At each timestamp, observed plant MW values are summed and integrated over a 15-minute interval. Missing readings are not silently certified as zero.

$$E(t)\ [\mathrm{MWh}] = 0.25 \times \sum_{q=1}^{96} \sum_{p} \mathrm{MW}(p,q)$$

A target day is eligible only with exactly 96 distinct intervals and no explicit null plant values. The two partial boundary days have 74 and 23 intervals and are excluded. Twelve additional days contain explicit missing readings. This leaves 815 eligible target dates. Reindexing to a complete daily calendar before constructing shifted histories prevents a missing day from being mistaken for a one-day lag. Removing rows without all required past values leaves 756 modelling rows; targets are never interpolated.

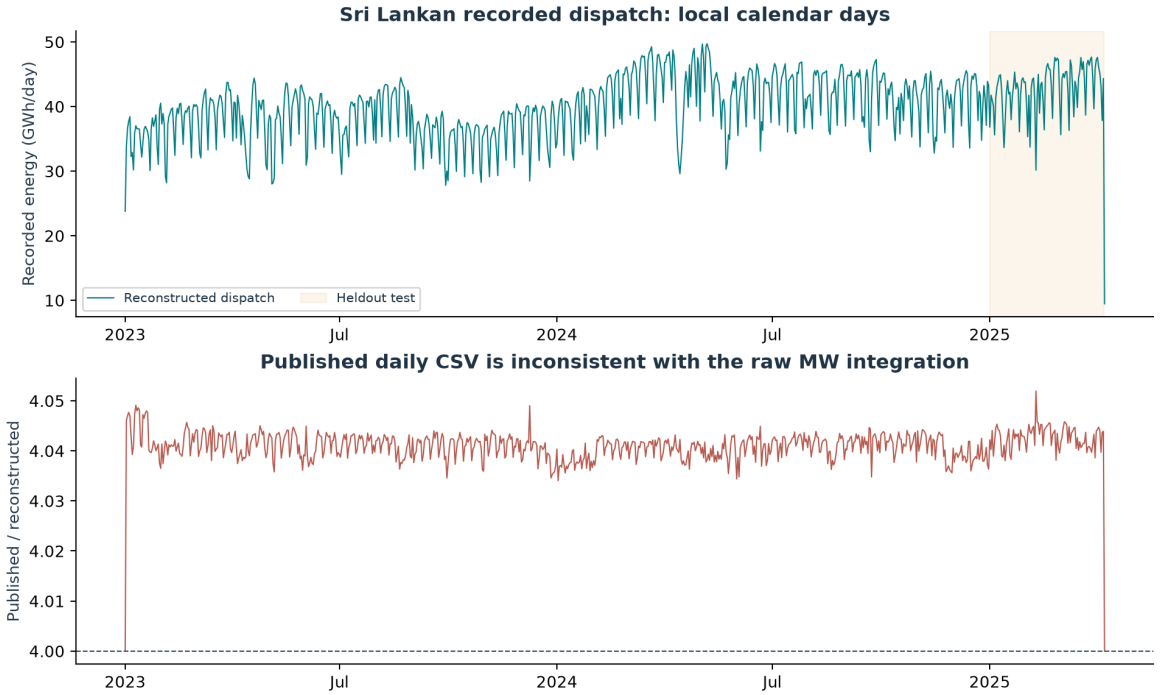


Figure 1. Reconstructed daily energy before eligibility filtering (top) and the published/reconstructed ratio (bottom). The low final point is a partial boundary day and is excluded from modelling.

The released daily CSV equals the local-day sum of raw, undeduplicated MW readings exactly (maximum absolute difference 0). Its median ratio to reconstructed MWh is 4.041427. This provides direct evidence that the published values omit the documented 0.25-hour conversion and include duplicate keys. Dividing that CSV by four would retain duplicate inflation. On 24 December 2024, reconstruction gives 41.668 GWh, compared with PUCSL's 41.96 GWh excluding non-telemetered generation and 44.92 GWh including estimates [9]. This supports the scale, while preserving a coverage caveat.

Small negative readings and ten all-zero system intervals are retained. Zero periods may represent interruption or telemetry failure; the values alone cannot resolve the cause. Clipping all negative readings would add 549.25 MWh over the full record, with a maximum daily change of 39 MWh. No extreme test target is removed to improve performance.

3. Why Gaussian Process Regression?

3.1 Evidence that the method is new to this course material

The supplied-material audit covers 38 files, including 1,214 PDF pages, 278 notebook cells, Office documents and a project slide image. Text extraction, OCR of low-text pages, binary-PowerPoint extraction and notebook-source review identify the taught model families. LightGBM and CatBoost are explicitly present; the Week 7 XAI note also discusses generalized additive models and explainable boosting machines. GPR is absent from the extracted/OCR material. Gaussian basis functions in linear regression and Gaussian mixture clustering are distinct techniques. This supports eligibility against the supplied evidence, while leaving unsupplied classroom discussion outside the audit.

3.2 Model and assumptions

GPR places a Gaussian prior over function values and conditions it on observations [4]. Similar standardized feature vectors have related outputs through a covariance kernel. This differs from fitting a fixed linear mean, partitioning feature space with trees, averaging forest predictions, or fitting the epsilon-insensitive objective used by SVR. Sharing a kernel family with SVM does not make the inferential objective or predictive distribution the same. GPR is appropriate here because the filtered daily table is small enough for exact inference and the study benefits from an uncertainty estimate.

$$f \sim GP(0, k);\ y = f(X) + \varepsilon;\ \varepsilon \sim N(0, \sigma^2_n I)$$

$$K_y = K(X,X) + \sigma^2_n I$$

$$\mu_* = k_*^T K_y^{-1} y$$

$$var(y_*) = k(x_*,x_*) - k_*^T K_y^{-1} k_* + \sigma^2_n$$

The implementation standardizes predictors using training data and normalizes the training target internally [5]. Cholesky solves are used instead of forming a matrix inverse explicitly. The equations show the normalized-space model; predictions are returned in MWh. A white-noise kernel represents observation noise, and a small diagonal jitter of 10^{-6} stabilizes the covariance matrix.

3.3 Kernel choice and tradeoffs

The search compares RBF, Matérn-3/2 and Matérn-5/2 signal kernels, each multiplied by a learned constant and added to white noise. A shared scalar length scale limits search complexity after feature scaling. Matérn-5/2 uses $k(r) = \sigma^2_f [1 + \sqrt{5}r/\ell + 5r^2/(3\ell^2)] \exp(-\sqrt{5}r/\ell)$. Smoothness, Gaussian noise, stationarity in transformed feature space and fixed hyperparameters remain assumptions, rather than established properties of the electricity system.

Exact GPR has cubic training cost and quadratic covariance storage in sample size. It becomes unattractive for millions of interval rows, motivating the daily task rather than indiscriminate full-resolution fitting. Its uncertainty is conditional on this kernel and fitted parameters; it does not automatically include structural shifts, missing plants or unknown publication delays.

4. Features and temporal validation

4.1 Forecast timing and leakage prevention

For target day t, predictions use only completed readings from t-1 or earlier and known calendar attributes of t. This assumes full previous-day telemetry is available at the forecast origin. Actual publication latency has not been verified, so the experiment is retrospective. During validation and testing, parameters remain fixed within each evaluation window, while yesterday's observed outcome becomes a permissible input for the next day. This is a rolling-input, one-day-ahead evaluation; it is not a multi-month forecast issued at one fixed origin.

Predictor group	Construction	Count
Demand history	E(t-1), E(t-2), E(t-7), E(t-14)	4
Recent level and variability	Mean and population SD over t-7 ... t-1	2
Weekly calendar	sin and cos of 2π × weekday/7	2
Annual cycle	sin and cos of 2π × (day-of-year - 1)/365.2425	2
Trend	Calendar days since 1 January 2023	1

There are no categorical raw predictors to encode. Cyclic transforms avoid treating Sunday and Monday as far apart. No same-day dispatch, contemporaneous weather, target-derived ratios or future rolling totals enter the predictors. Scaling is refitted inside every training fold, and test data never enter scaling, kernel fitting, model selection or interval recalibration.

4.2 Calendar-based expanding validation

Fold	Train through	Validation dates	Train / validation rows
1	31 Mar 2024	1 Apr-30 Jun 2024	395 / 80
2	30 Jun 2024	1 Jul-30 Sep 2024	475 / 92
3	30 Sep 2024	1 Oct-31 Dec 2024	567 / 92
Final	31 Dec 2024	Test: 1 Jan-7 Apr 2025	659 / 97

Chronological windows avoid training on future data, as cautioned by temporal validation guidance [6]. Custom date boundaries preserve comparable calendar seasons despite quality-related gaps; this is not an unqualified application of equally spaced TimeSeriesSplit. Each fold uses all eligible earlier rows, starting on 16 January 2023. Mean validation MAE across the three folds is the sole selection criterion. The 80-row first validation period receives the same fold weight as the two 92-row periods, a predeclared choice.

The final split is approximately 87.2% development and 12.8% test among eligible rows. Within development, the first validation fold begins after 395 training rows. A separate random train/validation/test split is inappropriate for this forecasting question. The experiment protocol and selected configurations are saved before test prediction.

5. Model selection and reproducibility

5.1 Fair and bounded comparisons

Two operationally transparent baselines predict yesterday's energy (persistence) and the same weekday last week (seasonal naïve). Taught comparators are ordinary linear regression, random forest and RBF SVR. All learned models see the same eleven features and evaluation dates. RF, SVR and GPR each have six candidate configurations; linear regression and the two naïve methods have no search. This controls search opportunity, although it does not guarantee equal representational power or equal optimization difficulty.

Model	Predeclared search and fixed choices
Linear regression	Intercept fitted; predictors standardized.
Random forest	300 trees; depth {6, unrestricted}; leaf minimum {2, 5, 10}; seed 42.
SVR	RBF kernel, gamma=scale; C {1,10,100}; epsilon {0.05,0.15}; X and y standardized within training.
GPR	RBF, Matérn-3/2 or Matérn-5/2; initial length {1,5}; learned signal amplitude, length and noise.

GPR maximizes training log marginal likelihood within each candidate using L-BFGS-B, limited to 80 iterations. Bounds are signal variance [0.01,100], length [0.1,100] and noise variance [0.0001,2] in normalized units. There are no random optimizer restarts in selection. The selected family is Matérn-5/2, initialized at length 5. Its final fitted signal variance is 4.834, length 8.228 and noise variance 0.185. No convergence warning is recorded in the candidate fits or selected final fits.

5.2 Selected validation results

Model	Mean validation MAE (GWh)	Selected settings
Random forest	1.823	Unrestricted depth; leaf minimum 5
GPR	1.998	Matérn-5/2; initial length 5
SVR	2.123	C=1; epsilon=0.05
Linear regression	2.182	No tuned hyperparameters
Persistence	2.911	Lag 1
Seasonal naive	3.025	Lag 7

Random forest is the best taught comparator selected by validation. GPR's selected-fold MAEs are 2.315, 2.187 and 1.492 GWh: substantial variation across evaluation periods. Differences between GPR initial lengths are tiny and should not be interpreted as a substantive advantage of the starting value. Complete candidate/fold scores are retained for audit.

6. Heldout predictive performance

MAE is primary because it expresses typical absolute error in the planning unit without giving a few extreme errors the quadratic weight of RMSE. RMSE is retained to expose large misses; R^2 measures squared-error skill relative to the test mean. Baseline skill is 1 - MAE(model)/MAE(seasonal naïve). Metric choice is fixed before evaluation; accuracy and F1 would be inappropriate for this continuous target.

Model	MAE (GWh)	RMSE (GWh)	R^2	MAE skill
Persistence	3.255	4.243	-0.323	-43.1%
Seasonal naive	2.274	3.173	0.260	0.0%
Linear regression	2.043	2.805	0.422	10.2%
Random forest	1.607	2.322	0.604	29.3%
SVR	1.534	2.406	0.574	32.5%
GPR	1.599	2.492	0.544	29.7%

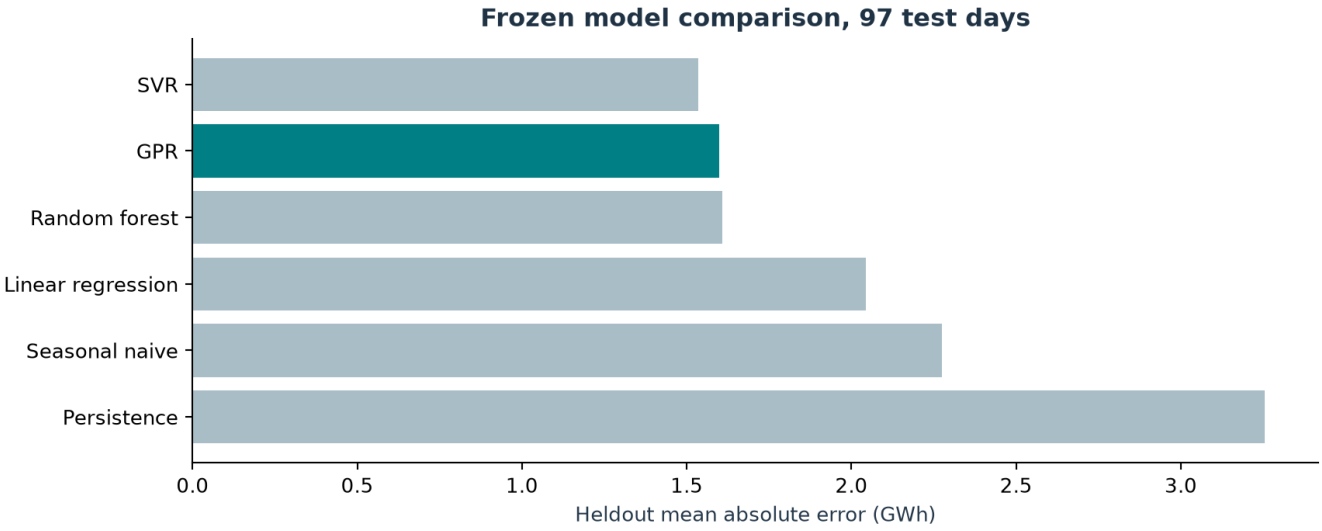


Figure 2. Mean absolute errors from frozen models on the same 97 test dates. GPR is highlighted as the new method, without implying that it is best on every metric.

GPR improves seasonal-naïve MAE by 0.675 GWh (29.7%) and persistence MAE by approximately 50.9%. SVR has the lowest test MAE, 1.534 GWh, whereas random forest has the lowest RMSE, 2.322 GWh. GPR's R^2 is 0.544: meaningful improvement over a constant test-mean reference, with considerable unexplained variation. The test does not establish that the new method dominates taught models.

Validation and test rankings differ: RF had the lowest validation MAE, but SVR has the lowest test MAE. This is reported descriptively and does not trigger retuning or replacement of the frozen GPR prototype. For point forecasting alone, RF remains the model preferred by the stated validation rule; GPR supplies a probabilistic alternative whose interval quality must also be examined.

7. Uncertainty and comparison stability

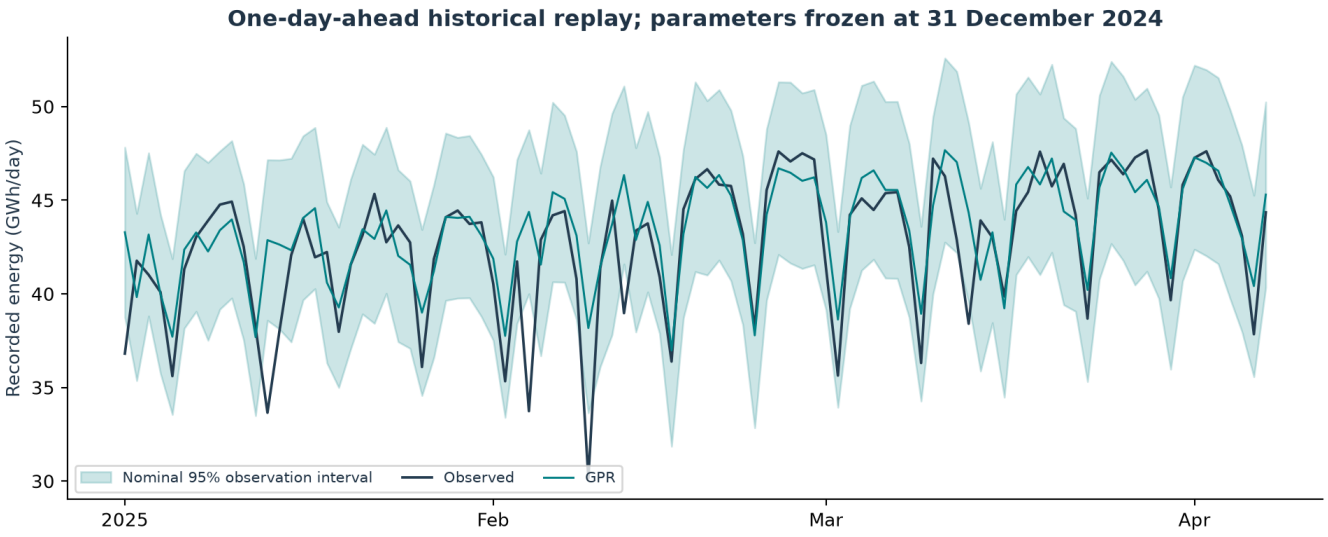


Figure 3. Observed and GPR-predicted dispatch energy with nominal 95% observation intervals. Each forecast uses only previous-day-or-earlier observations; parameters are not updated during the test.

The predictive intervals use $\mu \pm 1.960\sigma$, where σ includes the fitted white-noise observation component [5]. Coverage is 90/97 = 92.8%, compared with nominal 95%, and mean interval width is 9.340 GWh. Gaussian negative log predictive density is 9.274 using MWh as the target unit; this score changes with units and is useful only in like-for-like probabilistic comparisons. The broad intervals and seven misses argue against presenting the displayed range as an operational guarantee.

Observed coverage below nominal suggests a calibration concern, but 97 temporally related outcomes are insufficient to prove systematic undercoverage by themselves. The fitted Gaussian noise is homoscedastic, hyperparameter uncertainty is omitted, and known exceptional dates can shift the conditional mean. No test-based interval recalibration is performed. Prospective calibration and a wider evaluation period would be needed for decision support.

7.1 Paired moving-block bootstrap

Comparator	Mean GPR-minus-comparator MAE (GWh)	95% interval, 7-day blocks
Seasonal naïve	-0.675	[-1.134, -0.221]
Random forest	-0.009	[-0.188, +0.181]

Two thousand paired circular moving-block resamples preserve local dependence better than an independent-day bootstrap. Negative differences favour GPR. Fourteen-day-block sensitivity gives [-1.084,-0.257] GWh versus seasonal naïve and [-0.184,+0.168] GWh versus RF. Both block lengths support improvement over the weekly baseline; both include zero against RF. These intervals describe this short test under resampling assumptions and are not a multi-dataset significance claim.

RF seeds 7, 42 and 2026 produce test MAEs 1.631, 1.607 and 1.634 GWh. GPR initial-length multipliers 0.5, 1 and 2 converge to effectively identical fitted kernels and MAEs (difference below 0.002 MWh). These predeclared diagnostics are retained without choosing a better test-scoring run. GPR residual autocorrelations are -0.019 at lag 1 and -0.007 at lag 7; near-zero values do not prove independent errors.

8. Errors and Sri Lankan calendar effects

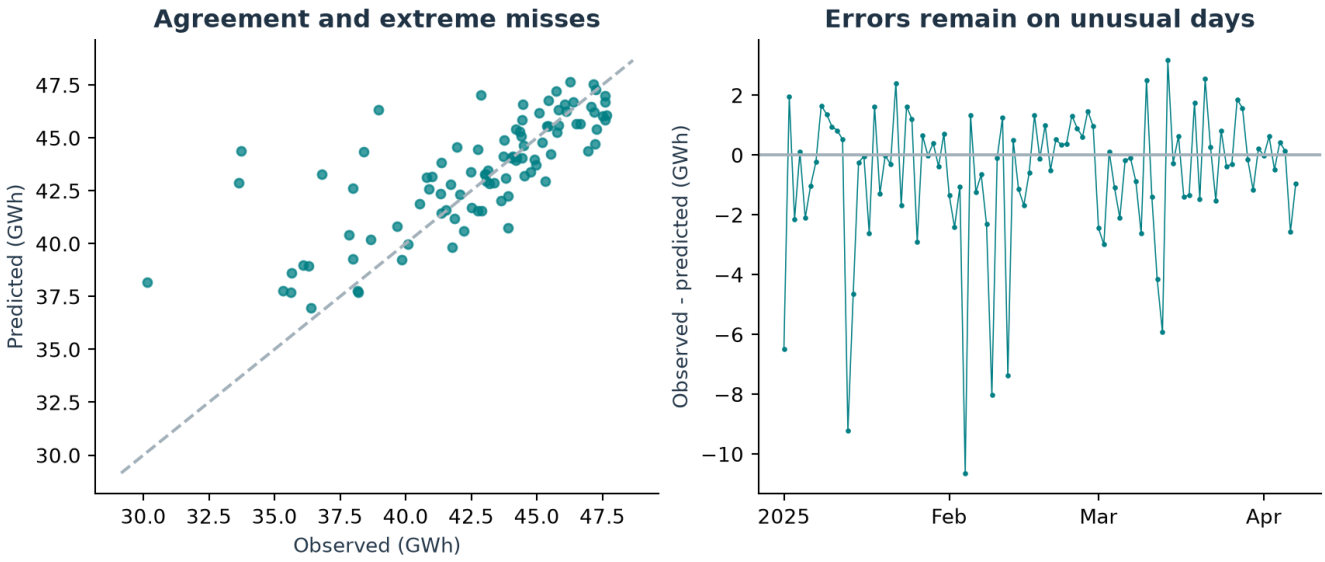


Figure 4. Agreement plot (left) and signed forecast residuals (right). Several unusual days have large overpredictions, consistent with omitted event information.

Date	Observed GWh	GPR GWh	Absolute error GWh
4 Feb 2025	33.735	44.373	10.638
13 Jan 2025	33.650	42.866	9.216
9 Feb 2025	30.152	38.173	8.021
12 Feb 2025	38.968	46.337	7.369

The largest miss occurs on 4 February, Independence Day. Another occurs on 13 January, Duruthu Full Moon Poya Day, and 12 February is Navam Full Moon Poya Day [10]. The model knows weekday and annual position, but has no explicit Sri Lankan holiday flag. These coincidences support a plausible omitted-calendar explanation; they do not establish a causal attribution or exclude concurrent changes in supply or telemetry. On 9 February, the raw record includes all-zero intervals; a low recorded outcome could reflect physical disruption or data issues.

January, February and March GPR MAEs are 1.657, 1.824 and 1.529 GWh, respectively. The April estimate is 0.746 GWh, based on only seven days; it should not be interpreted as a stable seasonal improvement. The heldout mean actual-minus-predicted residual is -0.683 GWh, indicating aggregate overprediction. Error slices and the ten worst predictions are provided as auditable tables.

Holiday engineering is a strong next experiment because the calendar is knowable in advance. It is deliberately not fitted after diagnosing these test errors. A revised holiday-aware model would require a newly reserved later holdout or a fresh outer validation design. Otherwise the same test would have served both feature selection and final assessment. The original difficult dates remain in every reported main comparison.

9. Global explanation and ablation

The official guide accepts feature importance as an explainability method [1]. Grouped permutation importance measures deterioration in heldout MAE when a predictor group is shuffled [7]. The six correlated demand-history variables are moved together; weekday sine/cosine and annual/trend coordinates form two further groups. Twenty shuffles rearrange contiguous seven-day blocks, then apply a random 0-6-day circular phase offset. The offset prevents identical weekday encodings after whole-week permutations.

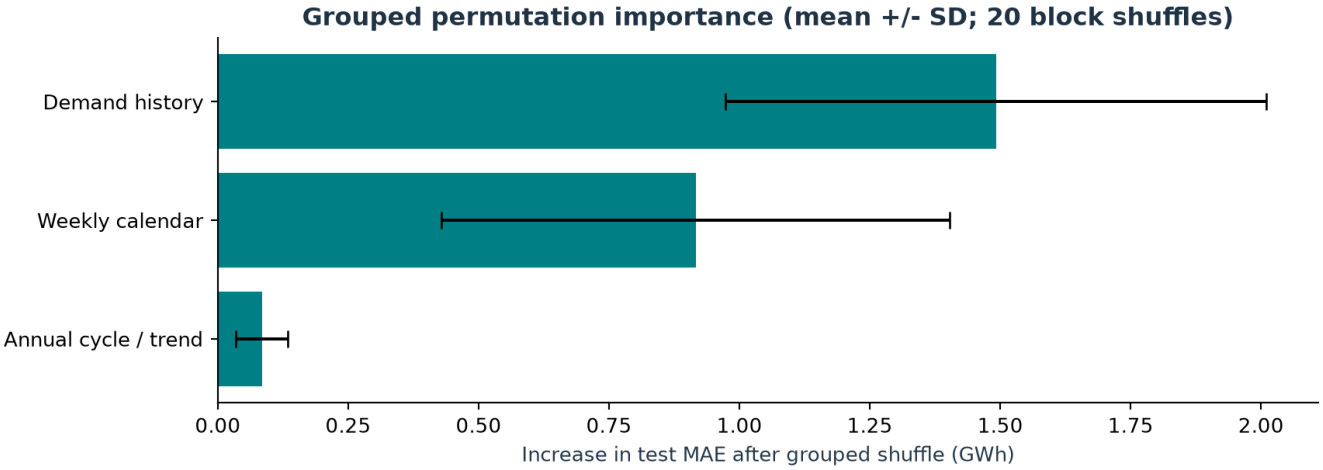


Figure 5. Mean increase in test MAE with standard deviation across 20 block shuffles. Error bars quantify shuffle variability, not uncertainty across future datasets.

Demand-history permutation raises MAE by 1.492 GWh on average (SD 0.518), weekly-calendar permutation by 0.916 GWh (SD 0.488), and annual/trend permutation by 0.085 GWh (SD 0.050). Thus the fitted GPR relies most on observed recent level and recurring weekly structure. This is consistent with the development data's lower weekend dispatch, rather than proof of any economic cause. Low importance for annual/trend features can reflect limited history, redundancy and the short test window, not their universal irrelevance.

9.1 Retraining asks a different question

GPR specification	Test MAE (GWh)	Change from full
Full eleven features	1.599	Reference
Demand history only	2.009	+0.411 GWh
Calendar and trend only	2.423	+0.824 GWh

Ablation models are refitted using the selected kernel family without retuning. Removing calendar/trend worsens MAE; removing history worsens it further. This complements permutation evidence: shuffling tests reliance of a fixed model, whereas retraining measures what a different model can recover from a restricted feature set. The limited search inherited by each ablation prevents claiming that these are optimal reduced-feature models.

Neither grouped shuffling nor retraining proves causality. Blocks retain some local structure but can still produce feature combinations outside the observed joint distribution. Group-level interpretation is preferable to splitting credit arbitrarily among redundant lags, and its limitations remain explicit [8].

10. Local effects and their limits

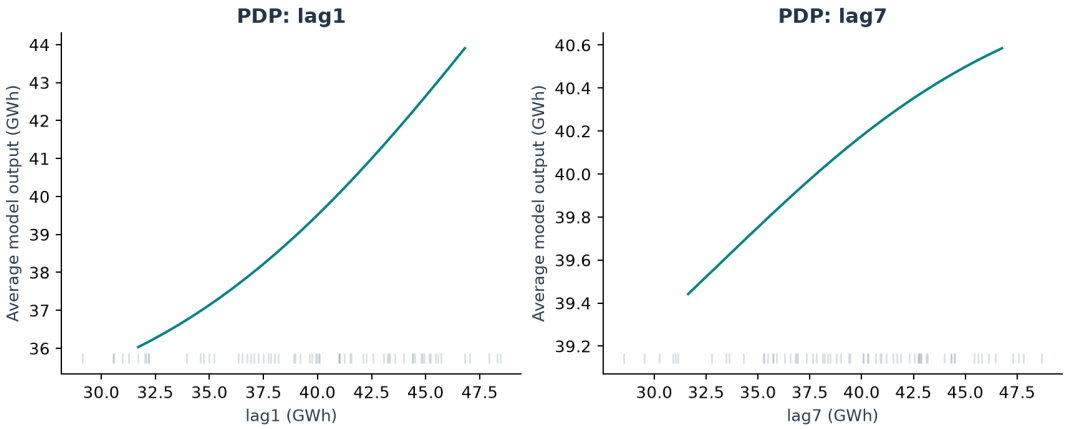


Figure 6. Marginal partial dependence over development-data reference rows; rug marks show observed feature support. Other lag variables are held to their reference values, so some combinations may be unrealistic.

PDPs average GPR predictions after replacing one feature over its development-data 5th-95th percentile range. Both previous-day and previous-week energy have positive average response curves. The displayed response to lag 1 is stronger over its plotted range than that to lag 7, but different ranges and correlated histories prevent interpreting a slope comparison as uniquely identified importance. PDP extrapolation concerns are particularly relevant when several lags describe the same physical process [8].

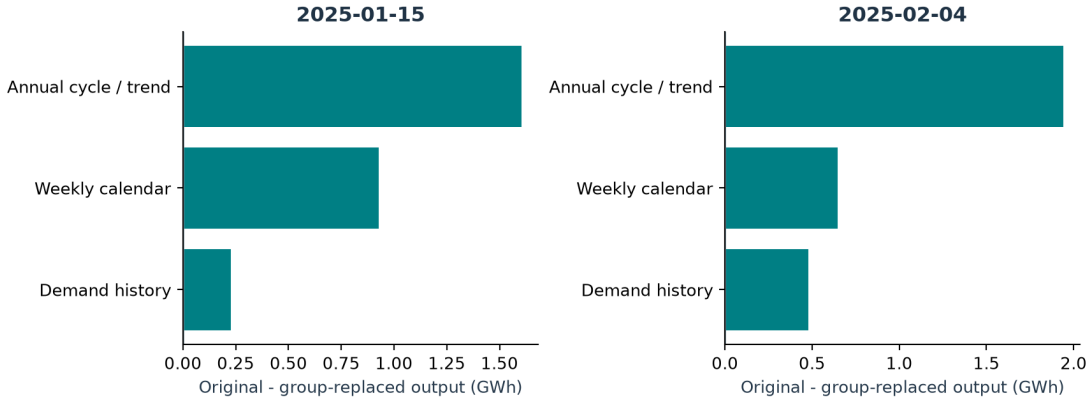


Figure 7. Non-additive group-replacement sensitivities for a typical replay and the largest-error day. The representative background is the actual development row for 12 January 2024.

The local reference is the actual development row nearest the standardized median vector, preserving valid cyclic coordinates within each reference group. For 15 January 2025, GPR predicts 42.326 GWh. Replacing demand history with the reference group lowers output by 0.226 GWh; replacing weekly calendar lowers it by 0.928 GWh; replacing annual/trend lowers it by 1.603 GWh. The same analysis is shown for 4 February, where the forecast is inaccurate.

These separate replacement differences do not sum to the forecast and are not SHAP values. Changing the reference changes the question, and cross-group hybrids can still be unsupported. A locally large annual/trend difference can coexist with low average permutation importance because the methods measure different quantities. Faithfully describing a mistaken forecast does not validate the forecast, establish fairness or provide causal recourse.

11. Critical discussion and practical impact

11.1 Data validity and representativeness

The dominant data risk is target meaning. Recorded plants vary from 33 to 42 at individual timestamps, and an absent plant is not a verified zero. Full timestamp coverage does not certify full national generation coverage. Dispatch responds to available supply and network conditions; it cannot independently reveal unsatisfied demand or household consumption. A model trained on this proxy should not be used to calculate household bills, attribute consumption to demographic groups or recommend dispatch set-points.

The raw values are integer-like MW readings with some negatives and zeros. Rectangular 15-minute integration follows the authors' documented procedure, but uncertainty about instantaneous versus interval-average reporting remains. Excluding explicit-null days protects target integrity while creating selection bias toward better-recorded dates. Dropping rows with unavailable histories may further underrepresent disturbed periods. The retained quality tables allow these exclusions to be inspected rather than hidden.

11.2 Validation and model limitations

Only a little over two years of data and one 97-day holdout are available. The test omits most of 2025, including Sinhala and Tamil New Year, and cannot establish robustness to future tariffs, new generating capacity, climate extremes or reporting changes. Three chronological folds are stronger than random shuffling but do not eliminate model-selection uncertainty. Six candidate settings per nonlinear model are a bounded search, not exhaustive optimization. Validation MAE is aggregated equally by fold despite different eligible counts.

GPR is a stationary, smooth model over standardized inputs, with scalar length scale and Gaussian homoscedastic noise. It can smooth abrupt holiday effects and extrapolate poorly as the trend coordinate moves beyond training support. Its probabilistic output adds information but also depends on assumptions that are visibly imperfect on exceptional dates. SVR and RF results demonstrate that algorithm novelty does not guarantee superior point accuracy. A simple, interpretable baseline remains essential to judge whether complexity is useful.

11.3 Bias, ethics and consequences

This national aggregate contains no protected attributes; consequently demographic parity or equalized-error claims cannot be measured. Coverage bias may nevertheless reflect where reporting plants and distributed generation are represented. National accuracy can conceal regional shortages and different household experiences. Overprediction could encourage unnecessary energy procurement; underprediction could weaken planning margins. These are potential pathways, not quantified savings or harms demonstrated here. Human operational expertise, verified live inputs and prospective monitoring would be prerequisites for real deployment.

11.4 Next evidence to collect

A future study should add advance-known Sri Lankan holidays, verify telemetry publication latency and use issued weather forecasts. A longer untouched period should test calibration, seasonal stability and reporting changes. Conditional feature-effect analyses could strengthen XAI. Local compute handles training; the Azure review deployment demonstrates reproducible inference without establishing operational readiness.

12. Prototype and reproducible deliverables

12.1 An interface connected to the actual fitted model

The “Lanka Dispatch / Forecast Lab” interface accepts a target date and six demand-history summaries, using GWh for readability. Calendar features are derived from the selected date. A Python HTTP service loads the saved GPR pipeline and returns the actual model mean, observation interval, seasonal-naïve comparison and signed group-replacement sensitivities. The browser contains no mock prediction function. Recorded outcomes are displayed separately for historical replay; editing inputs never rewrites the observed outcome.

Users can load a historical case, edit values, generate a fresh estimate, inspect evaluation evidence, and read the source/quality notes. Negative, non-finite and out-of-range energy inputs are rejected. Out-of-development energy ranges and dates outside the evaluated period generate warnings. The original development server binds to localhost; a separate Flask/Gunicorn wrapper supports the verified HTTPS Azure review deployment described in Section 17. Credentials remain outside the source package. A captured-interface walkthrough accompanies the source; its narration is synthesized and disclosed as assistance.

12.2 Reproduction workflow

Stage	Entry point / evidence
Acquire and prepare	03_code/prepare_data.py; raw hashes and daily quality table
Select and evaluate	03_code/train_evaluate.py; protocol, complete CV log, saved models and test predictions
Explain and plot	03_code/explain_and_plot.py; grouped shuffle repeats, PDP data, local sensitivities and figures
Verify	03_code/validate_project.py; independent lag and metric checks
Demonstrate	03_code/app.py; local browser interface and API checks
Rebuild report	03_code/build_report.py; editable content JSON and Markdown source

A notebook runs the same canonical scripts so that its results do not diverge from a separately maintained implementation. The README specifies environment creation, pinned dependencies and launch commands. The source ZIP includes permitted Sri Lankan data, processed tables, source, notebook, model artefacts, logs and explanations, but excludes the virtual environment, other countries’ data, copied teaching materials, credentials and scratch files. The separate PDF and 3-5 minute video satisfy the guide’s listed submission formats [1].

Verification checks independently reconstruct lags and rolling histories, enforce chronological separation, confirm development-only scaling and GP training size, reproduce saved predictions, recompute every main metric, and confirm validation-only candidate selection. An isolated reproduction checks the packaged workflow. Student review, applicable AI rules and Moodle upload remain human actions.

13. Extension design and NGBoost

The extension responds to a request for broader evidence rather than a change in the primary research question. The original study, selected kernel, saved model and test outcomes remain unchanged. Its test had already been inspected: NGBoost, KNN and gradient boosting results on electricity are therefore **supplementary post-hoc comparisons**. Their settings still use the original three development folds; the extension must not be described as a newly untouched electricity test. The initial rainfall protocol preceded scoring. A publication-timing review then found unavailable year-end training labels. Version 1.1 removes the six dekads before each January evaluation origin from training. Invalid preliminary results are archived. Since outcomes had already been inspected, corrected rainfall scores are retrospective reanalysis, not newly untouched testing; no score-driven tuning followed.

13.1 A second unfamiliar probabilistic algorithm

NGBoost models conditional distribution parameters rather than only a point prediction. Here the Normal distribution has a mean and log-scale. A proper logarithmic score evaluates the full predictive distribution. The natural gradient preconditions score derivatives by the distribution's information geometry, helping coordinate parameters measured on different scales [13]. Weak regression trees fit these adjusted directions, followed by a shared step-size update. This differs from conventional squared-error gradient boosting, which directly targets the conditional mean.

Natural gradient: g_nat(theta) = I(theta)^(-1) grad_theta S(theta, y)

Normal prediction: Y | x ~ Normal(mu(x), sigma(x)^2)

This project uses the maintained NGBoost implementation, version 0.5.11, Normal/LogScore and natural_gradient=True. Target mean and standard deviation are fitted inside each training fold and reversed for prediction. Six fixed candidates combine tree depth 2/3 with 200/400/600 boosting rounds, learning rate 0.03 and minimum leaf size 10. Negative rainfall support under a Normal family is explicitly checked rather than concealed by clipping.

13.2 Fair search and course alignment

KNN uses training-only standardization, 5/15/31 neighbours and uniform/distance weights. Conventional gradient boosting uses depths 1/2 and 100/300/600 rounds, learning rate 0.03 and minimum leaf size 10. Each new nonlinear model receives six candidates and three expanding folds; seed 42 is fixed. Rainfall also reuses the original LR, RF, SVR and GPR search definitions. NGBoost and natural-gradient boosting were not found in the supplied lecture extraction; GPR remains the assignment's focal new method. No classification, deep-learning or image-processing application is introduced.

The frozen extension protocol records the snapshot hash, features, folds, candidates and reporting rules. Complete candidate scores, selected-before-test settings, fitted objects, point/probabilistic metrics and execution logs are supplied. Two runs reproduced all extension predictions within 10^-8 in native target units; 78 independent checks passed.

14. Expanded electricity comparison

Table 6 retains every original result and adds the three predeclared extension models. The unit is GWh; all learned models use the same 659 development and 97 test days. Original and added comparisons have different test-inspection histories and are labelled accordingly. Point predictions from probabilistic models are their conditional means.

Model	Status	Test MAE	Test RMSE
SVR	Original	1.534	2.406
GPR	Original	1.599	2.492
Random forest	Original	1.607	2.322
NGBoost	Added post-hoc	1.623	2.338
Gradient boosting	Added post-hoc	1.641	2.363
KNN	Added post-hoc	1.774	2.458
Linear regression	Original	2.043	2.805
Seasonal naive	Original	2.274	3.173
Persistence	Original	3.255	4.243

Table 6. Expanded dispatch errors, GWh. Exact unrounded metrics and configurations are in the experiment CSVs. Added models do not replace the original frozen comparison.

14.1 What the added evidence changes

NGBoost obtains MAE 1.623 GWh and RMSE 2.338 GWh, close to RF on this period. KNN reaches MAE 1.774 GWh; gradient boosting reaches 1.641 GWh. None improves on SVR's test MAE of 1.534 GWh or RF's RMSE of 2.322 GWh. Gradient boosting has the lowest validation MAE in the expanded comparison (1.714 GWh), while RF was the best original validation comparator (1.823 GWh). These scopes must not be conflated. Expanded-CV success does not guarantee the best heldout accuracy.

14.2 Distributional quality matters

NGBoost's nominal 95% Normal intervals cover 87.6% of electricity outcomes, with mean width 6.224 GWh and Gaussian negative log predictive density (NLPD) 9.622. GPR covers 92.8%, width 9.340 GWh and NLPD 9.274. Lower NLPD is better within the same target units; NGBoost's narrower intervals lose coverage. Neither nominal level is guaranteed, and cross-dataset NLPD values in different units should not be compared.

The evidence supports GPR as a defensible probabilistic research method, not the universal accuracy winner. NGBoost adds a useful alternative and an instructive calibration limitation. The deployed model stays the original GPR so that the demonstration matches the main XAI and uncertainty analyses. No test-directed retuning or holiday-feature insertion is performed.

15. Independent Colombo rainfall task

To test whether conclusions depend on the electricity domain, models are refitted on WFP/HDX Sri Lanka CHIRPS-v2 dekadal rainfall [14, 15]. This is a separate tabular task, not external validation of the electricity-trained model. The source combines satellite and gauge information into existing administrative aggregates; this project neither processes images nor collects original measurements.

Item	Prespecified choice
Target and locality	Final rfh rainfall, mm per dekad; Colombo district, admin level 2, PCODE LK11
Snapshot	56,066 country-wide rows; 25 districts + 9 provinces; 1,647 final Colombo records; pinned CSV and metadata retained
Model period / split	936 rows, 2000–2025; 858 development (Jan. 2000–Oct. 2023), six embargoed, 72 test (2024–2025)
Validation	Train through Oct. 2020 / validate 2021; then Oct. 2021 / 2022; Oct. 2022 / 2023. No shuffled splitting.
Predictors	Lags 6, 7, 9 and 36 dekads; three-dekad mean/std ending at lag6; annual sine/cosine; trend
Baselines	Prior-year same-dekad value; training-only median for each annual dekad
Quality	Final-only, nonnegative finite rainfall; complete calendar before shifting; no interpolation or extreme trimming

15.1 Availability and leakage controls

A year contains 36 dekads beginning on days 1, 11 and 21; the final period of a month has variable length. Features are built on this full calendar. WFP metadata states that final observations are published mid-month after the observed month. A minimum six-dekad input delay conservatively handles final-product latency. Training targets also stop at the preceding October, excluding six November–December dekads before each January evaluation origin. Archived issuance vintages are unavailable, so later revisions remain a retrospective limitation. The target is a future completed dekad relative to the forecast origin, not observed rainfall passed as a predictor.

Same-period rolling totals and anomaly ratios are excluded. The supplied climatology is also excluded; the baseline median is recalculated from training rows in each fold and only targets through October 2023 for the final test. Every learned transform uses training data. A calendar lookup independently verifies all lag values. Missing targets are never imputed.

The rolling HDX resource may change. Its SHA-256 is pinned in the protocol and acquisition code; the original snapshot is included in the repository and source ZIP. A changed future download is rejected instead of silently changing the experiment. HDX metadata specifies Creative Commons Attribution International; it does not identify a numbered CC version. Full attribution and the original licence metadata are preserved.

16. Rainfall results and negative findings

Model	CV MAE / mm	Test MAE / mm	Test RMSE / mm
Seasonal median	52.26	41.39	58.69
SVR	54.30	48.17	62.83
KNN	52.09	49.47	63.41
GPR	53.25	51.86	65.49
Random forest	51.52	52.28	64.64
Linear regression	53.99	53.13	66.17
Gradient boosting	54.61	55.72	67.36
NGBoost	53.68	63.09	75.18
Seasonal naive	67.52	66.89	87.92

Table 7. Independent rainfall comparison. RF has the lowest validation MAE (51.52 mm), while a seasonal median has the lowest heldout error (41.39 mm). SVR is the best learned model on test MAE (48.17 mm). Choosing the test winner for future use would require a new validation study.

GPR test MAE is 51.86 mm, exceeding the seasonal median by 10.46 mm. A 2,000-replicate paired circular block bootstrap gives a 95% difference interval [2.31, 19.17] mm for three-dekad blocks and [1.37, 20.92] for six-dekad blocks. With only two test years, these estimates depend on temporal assumptions. The SVR-versus-median six-dekad interval crosses zero; descriptive ordering should not become a universal superiority claim.

GPR’s nominal 95% rainfall intervals cover 94.4% but average 257.91 mm wide; 51/72 lower bounds are negative. NGBoost covers only 80.6%, width 196.75 mm, and has four negative lower bounds. Both produce nonnegative test means. Gaussian support and heteroscedastic extremes are limitations, not repaired by silently clipping intervals. A positive-support distribution, weather forecasts or revised features are future hypotheses needing fresh evaluation.

This negative finding strengthens the critical conclusion: unfamiliar probabilistic methods can be useful without consistently beating well-matched seasonal baselines. Raw MAE and NLPD across electricity and rainfall have different units and are not averaged into a single ranking.

16.1 Visual comparison across tasks

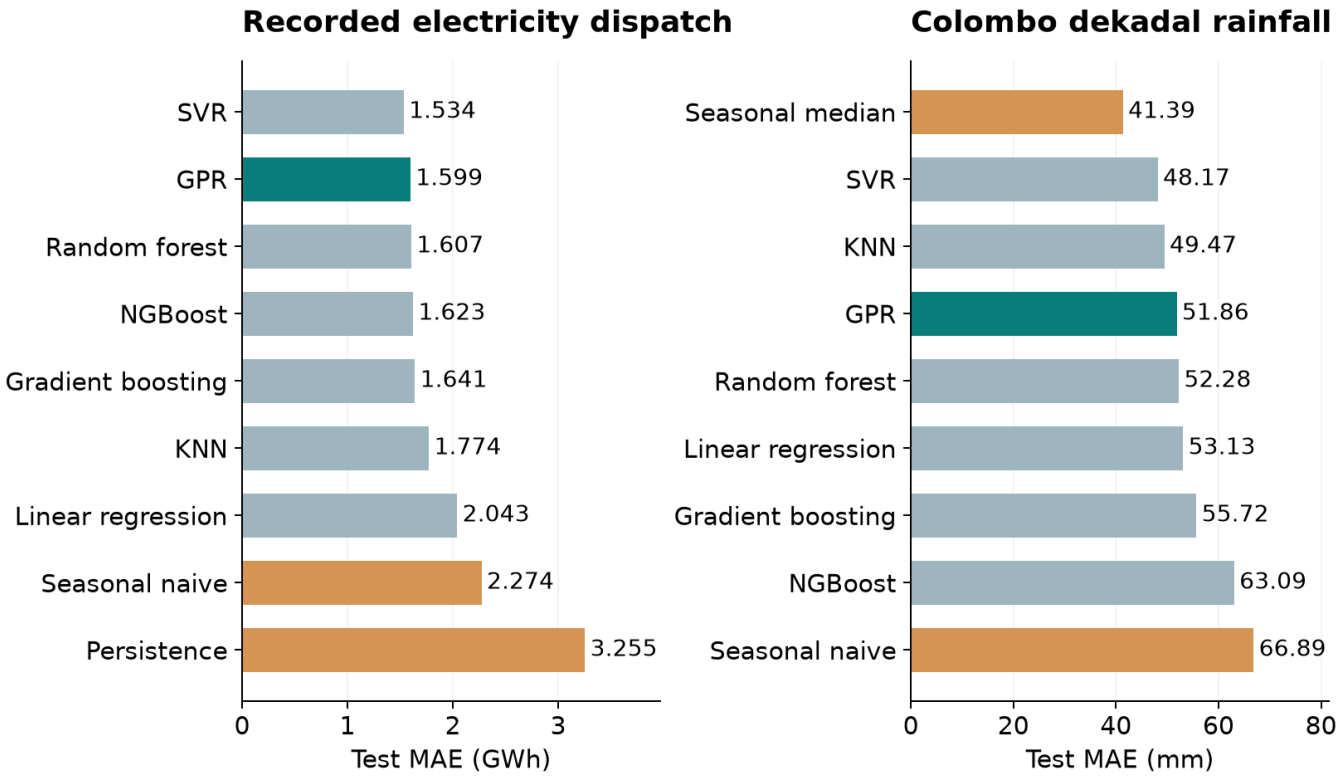


Figure 10. Full original and supplementary comparison on electricity (left) and corrected rainfall audit (right). Lower MAE is better. Separate units and baselines prevent a meaningful raw-MAE average across tasks.

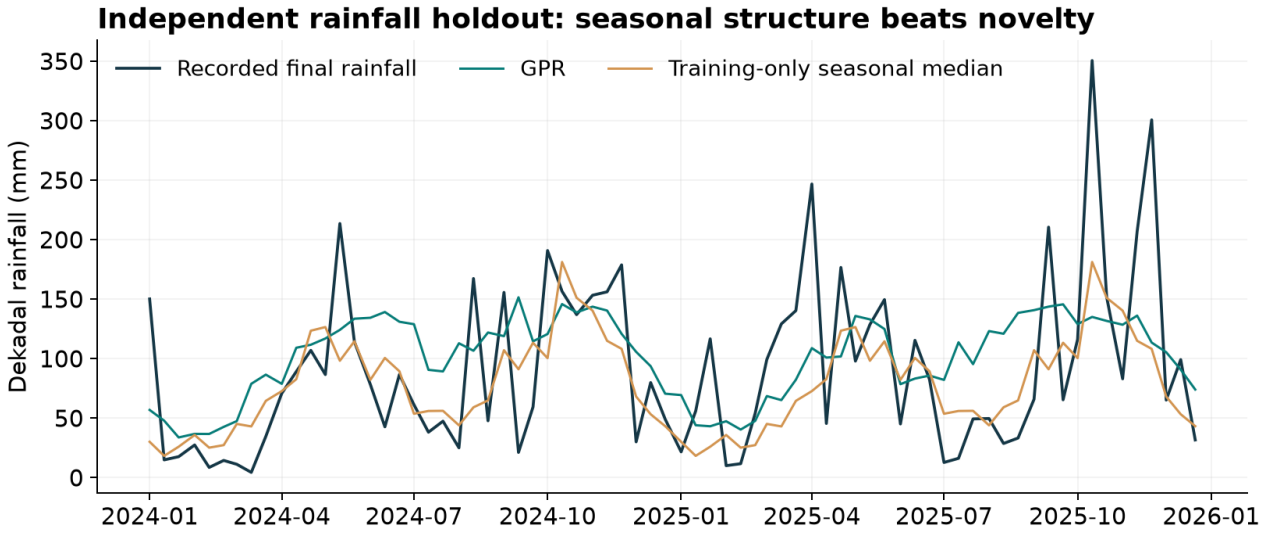


Figure 11. Actual final rainfall and two frozen predictions on 72 heldout dekads. Peaks remain difficult for the learned mean model.

The electricity plot preserves the original comparison and labels the extension’s status in Section 14. The rainfall plot uses training targets only through October 2023; six late-year dekads are embargoed before the January 2024 origin. Seasonal structure remains useful while peaks challenge mean forecasts. These are recorded results, not evidence that the trained electricity model transfers to rainfall.

17. Repository and Azure deployment

The requested review deployment runs the original saved GPR through a Flask/Gunicorn WSGI wrapper. The same app serves the interface, model API, original evaluation evidence and corrected extension tables. A private GitHub repository contains source, protocols, data instructions and result artefacts. The separate source ZIP remains an accessible assessment alternative when the reviewer has not been granted private-repository access.

Item	Verified configuration / evidence
Live HTTPS interface	https://lanka-dispatch-it5514-sv.azurewebsites.net
Private source repository	https://github.com/Sandaru-Vithanage/it5514-sri-lanka-ml
Hosting	Azure App Service Linux, East Asia; Python 3.13; one Gunicorn worker, two threads
Plan and cost boundary	F1 / Free, one app. User-approved maximum US$10 of student credit; no paid resources created.
Release provenance	Application source commit 79d46d9f2c74; exact model SHA-256 is exposed by /api/health and saved in deployment_record.json.
Remote validation	11 API checks passed, including identical model bytes, forecast agreement, invalid requests and equality of published extension results.

17.1 What deployment establishes

The HTTPS service returns 42,325.975574 MWh for recorded 15 January 2025 inputs and 44,217.785362 MWh after changing only lag1 to 42,000 MWh. Both match local inference within 10^-6 MWh; the historical actual remains fixed. Browser review verifies these interactions, the last-valid-estimate label, original and extended evidence, and the displayed URL, private source link, source commit and model fingerprint. The 3-5 minute video shows genuine hosted interface captures with disclosed editing and synthesized narration.

An explicit 14-file bundle excludes raw archives, notebooks, lecture materials, other model pickles and credentials. Per-file hashes and a ZIP checksum are recorded. Runtime dependencies are pinned separately from training/authoring packages. HTTPS-only service, TLS 1.2 minimum and disabled FTP are configured. Azure authentication state stays in an excluded local directory; no password, access token or subscription identifier is included in deliverables.

17.2 Cost, access and operational limits

The selected F1 tier is free shared compute with CPU quotas and cold starts [16]. The verified resource inventory contains only the free plan and app. This records a free configuration, not a fabricated final invoice or hard budget-enforcement mechanism. No paid monitoring, GPU, Azure ML endpoint, database, registry or custom domain was provisioned. Cleanup instructions are supplied for the dedicated resource group after assessment.

Hosting demonstrates reproducible inference, not real-time data availability, an operational SLA or grid suitability. The model was not retrained on the heldout period. The repository remains private by the student’s instruction; marker access requires a deliberate invitation or use of the submitted source ZIP. Durable video and source evidence remain available if free-tier quotas or cold starts interrupt the site.

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Online sources were checked on 28 October 2026. The verified source register records URLs, dates, source role and relevant evidence. Supplied lecture notes and the personal open-book guide inform course-coverage checking; their exercises do not replace the official assessment brief. No unverified source DOI or citation placeholder is used.

Assistance, review and student responsibility

AI assistance disclosure

OpenAI Codex assisted with reading and extracting the supplied materials, web research and source verification, dataset acquisition, experimental design, Python implementation, local execution, the extended rainfall/NGBoost experiments, result checking, visualizations, report drafting, private GitHub repository preparation, Azure deployment, the interface and demo preparation. The code was run on the student's MacBook; reported scores come from the saved experimental outputs. The demo uses a synthesized system voice and captured states of the working interface. These contributions are disclosed so that the work is not represented as unaided authorship.

The dataset authors and original producer retain credit for data collection. Source scripts were consulted for provenance, while this project implements its own reconstruction and modelling workflow. No claim is made that the student personally collected original measurements, independently authored every sentence or manually performed each execution step.

Academic-integrity boundary

The supplied brief and notes do not state an AI-specific assessment rule, and the student reports no additional instructions. University of Moratuwa's published plagiarism guidance requires full acknowledgement of others' work [12]. Absence of an AI-specific statement in the supplied material is not evidence of blanket permission. This draft requires student review and compliance with any programme, module or submission declaration. No independent-work declaration has been signed or fabricated.

What the student must understand before submission

The student should be able to explain why MW must be integrated into MWh; demonstrate the raw-data discrepancy; distinguish recorded dispatch from end-use demand; reconstruct lagged features by date; describe the GP posterior and Matérn kernel; explain why scaling is confined to each training fold; and distinguish a rolling-input forecast from a fixed-origin multi-step forecast.

The student should also explain the NGBoost natural gradient, rainfall publication delay, post-hoc status of added electricity comparisons, and why a seasonal median wins the rainfall task. The student should defend the use of temporal baselines, the validation-only selection rule, the absence of a GPR superiority claim over RF, the meaning and limitations of 92.8% interval coverage, the difference between permutation and retrained ablation, and why local group sensitivities are non-additive and non-causal. The accompanying defence guide provides focused questions and code locations for practice.

Remaining human actions

Read and revise this draft in the student's own understanding; confirm the lecturer's acceptance of GPR as a method outside supplied lecture coverage and any AI-use declaration; check the precise Moodle cutoff on 1 November 2026 and upload limits; review the demo and decide whether to rerecord personal narration; and upload the verified submission files. This package does not claim that Moodle upload, lecturer approval or an institutional plagiarism/AI review has occurred.

The completed technical checks and their evidence are recorded separately from these human actions. No fabricated score, execution log, consent record, deployment or approval is included.